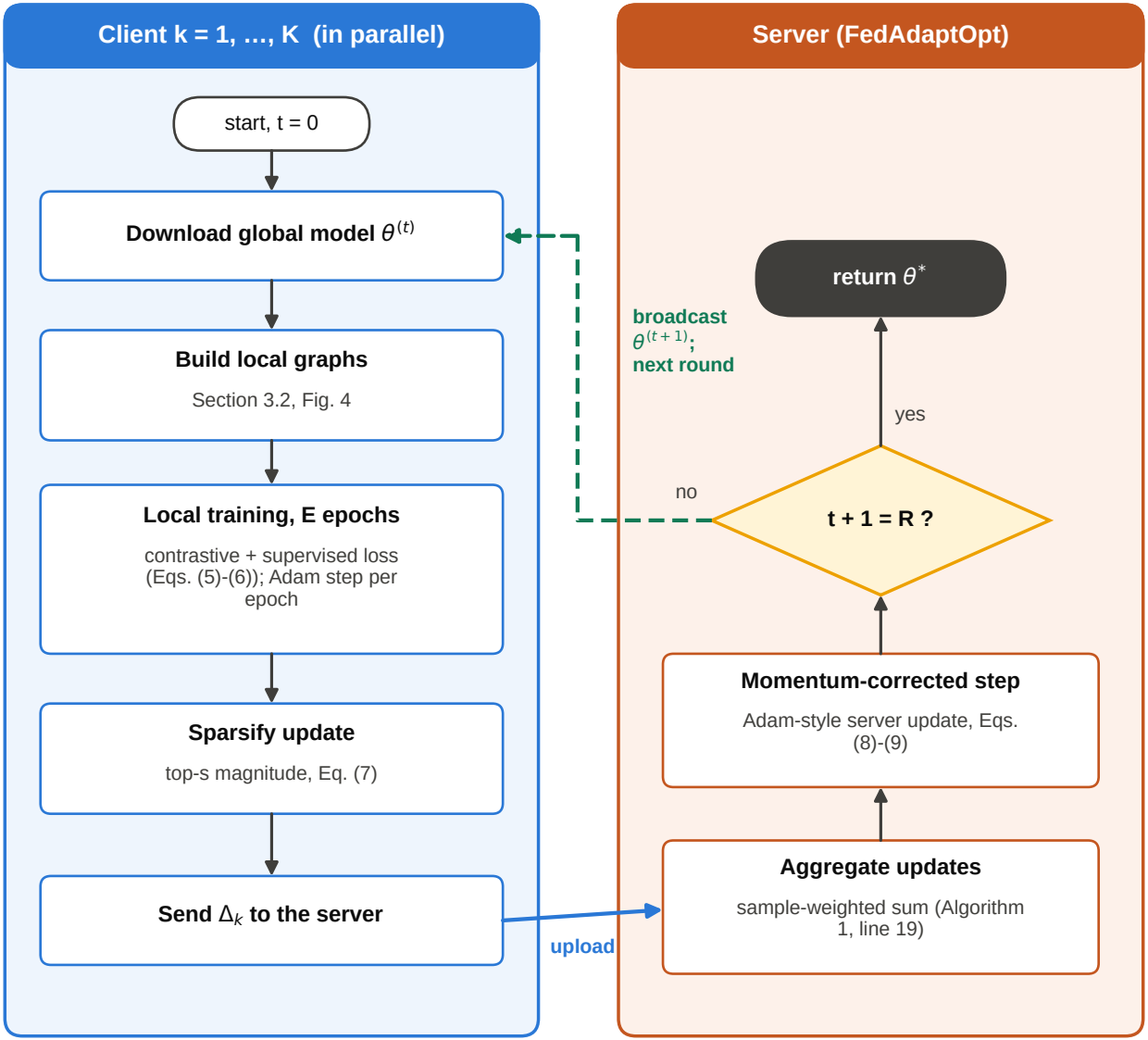
Client k = 1, …, K (in parallel)
Server (FedAdaptOpt)
start, t = 0
Download global model $\theta^{(t)}$
Build local graphs
Section 3.2, Fig. 4
Local training, E epochs
contrastive + supervised loss (Eqs. (5)-(6)); Adam step per epoch
Sparsify update
top-s magnitude, Eq. (7)
Send Δ_k to the server
upload
Aggregate updates
sample-weighted sum (Algorithm 1, line 19)
Momentum-corrected step
Adam-style server update, Eqs. (8)-(9)
t + 1 = R ?
yes
no
return θ^*
broadcast $\theta^{(t+1)}$; next round